

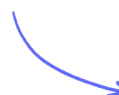
## Non-Calculator Questions

# Powers, Roots & Standard Form

Powers & Roots / Laws of Indices / Converting to & from Standard Form /  
Operations with Standard Form

|                      |            |
|----------------------|------------|
| Easy (12 questions)  | /14        |
| Medium (7 questions) | /14        |
| Hard (7 questions)   | /19        |
| <b>Total Marks</b>   | <b>/47</b> |

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# Easy Questions

- 1 The Earth has a surface area of approximately  $510\,100\,000\text{ km}^2$ .

Write this surface area in standard form.

.....  $\text{km}^2$

**(1 mark)**

- 2 Write 15 060 in standard form.

**(1 mark)**

- 3 Write each number in standard form.

i) 72 000

[1]

ii) 0.0018

[1]

**(2 marks)**

- 4 Write 0.00527 in standard form.

**(1 mark)**

- 5 Write 2 760 000 in standard form.

**(1 mark)**

- 6 Write 23 000 in standard form.

(1 mark)

7 Write 0.00813 in standard form.

(1 mark)

8 Write 0.000 038 7 in standard form.

(1 mark)

9 Here are some numbers written in standard form

$$3.4 \times 10^{-1} \quad 1.36 \times 10^6 \quad 7.9 \times 10^0 \quad 2.4 \times 10^5 \quad 5.21 \times 10^{-3} \quad 4.3 \times 10^{-2}$$

From these numbers, write down;

i) the largest number,

[1]

ii) the smallest number.

[1]

(2 marks)

10 Write  $4.82 \times 10^{-3}$  as an ordinary number.

(1 mark)

11 Write 52 million in standard form.

(1 mark)

12 Write 0.000 052 3 in standard form.

(1 mark)

# Medium Questions

- 1 Work out, giving your answer in standard form,

$$(5.2 \times 10^7) + (5.2 \times 10^6).$$

(2 marks)

- 2 Work out  $(3 \times 10^{199}) + (2 \times 10^{201})$ .

Give your answer in standard form.

(2 marks)

- 3 Work out  $(6.4 \times 10^7) + (9.6 \times 10^6)$ .

Give your answer in standard form.

(2 marks)

- 4 Write the following numbers in order of size. Start with the smallest number.

$$0.038 \times 10^2 \quad 3800 \times 10^{-4} \quad 380 \quad 0.38 \times 10^{-1}$$

(2 marks)

- 5 Work out the value of  $(9 \times 10^{-4}) \times (3 \times 10^7)$

Give your answer in standard form.

(2 marks)

- 6 Patrick has to work out the exact value of  $64^{\frac{1}{4}}$

Patrick says,

" $\frac{1}{4}$  of 64 is 16 so  $64^{\frac{1}{4}} = 16$ "

Explain what is wrong with what Patrick says.

(1 mark)

- 7 Evaluate.

$$\frac{(3 \times 10^{-3}) \times (8 \times 10^5)}{(5 \times 10^{-4}) \times (6 \times 10^{-2}) \times 2}$$

Write your answer in standard form.

(3 marks)

# Hard Questions

1 Find the value of  $64^{\frac{2}{3}}$ .

(2 marks)

2 Calculate  $0.125^{-\frac{2}{3}}$ .

(3 marks)

3 Work out.

$$\left(\frac{125}{27}\right)^{-\frac{2}{3}}$$

(3 marks)

4 Find the value of  $\left(\frac{1}{81}\right)^{-\frac{3}{4}}$ .

(2 marks)

5 (a) Write 5400 000 as a number in standard form.

(1 mark)

- (b) Write  $3.2 \times 10^{-4}$  as an ordinary number.

(1 mark)

- (c) The mass of the Sun is  $2 \times 10^{30}$  kg.  
The mass of the largest known star is 315 times the mass of the Sun.

Work out the mass of this star.  
Give your answer in kg in standard form.

(2 marks)

6 Simplify  $8^2 \times \sqrt[3]{4^6}$

Give your answer in the form  $2^a$  where  $a$  is an integer.

Show each stage of your working clearly.

(3 marks)

- 7 Work out, giving your answer in standard form,

$$(7.1 \times 10^{-15}) \times (2 \times 10^3)$$

(2 marks)